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Abstract

Achieving ultra-reliable communication under strict end-to-end latency constraints in integrated 5G-TSN systems requires fault-tolerant mechanisms that extend beyond a singleleg wireless transmission. Existing approaches that extend IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER) to 5G through redundant PDU sessions or dual connectivity, remain restricted to two-leg redundancy, lack correlation awareness, and operate under static replication policies. This paper proposes M-FRER, a multi-connectivity extension of FRER that enables replication and elimination across M heterogeneous connectivity legs, including multi-RAT, multi-PDU sessions, and non-3GPP access. M-FRER introduces (i) a correlation-aware reliability model that captures shared-risk dependencies between legs, and (ii) an adaptive replication controller that selects the active replication set while minimizing bandwidth, energy, and control overhead. Evaluations combining trace-based latency modeling with analytical reliability bounds under correlation show that M-FRER achieves five-nines on-time reliability ([?] 99.999%) with only modest additional bandwidth compared to dual-connectivity replication, and maintains bounded jitter through controlled elimination windows. These results indicate that deadline-compliant communication over stochastic wireless media is achievable when redundancy and control are jointly optimized, positioning M-FRER as a scalable foundation for TSNintegrated industrial 5G deployments.

M-FRER: A Multi-Connectivity Framework for Reliable and Deterministic 5G-TSN Integration

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Abstract—Achieving ultra-reliable communication under strict end-to-end latency constraints in integrated 5G-TSN systems requires fault-tolerant mechanisms that extend beyond a single-leg wireless transmission. Existing approaches that extend IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER) to 5G through redundant PDU sessions or dual connectivity, remain restricted to two-leg redundancy, lack correlation awareness, and operate under static replication policies. This paper proposes M-FRER, a multi-connectivity extension of FRER that enables replication and elimination across M heterogeneous connectivity legs, including multi-RAT, multi-PDU sessions, and non-3GPP access. M-FRER introduces (i) a correlation-aware reliability model that captures shared-risk dependencies between legs, and (ii) an adaptive replication controller that selects the active replication set while minimizing bandwidth, energy, and control overhead. Evaluations combining trace-based latency modeling with analytical reliability bounds under correlation show that M-FRER achieves five-nines on-time reliability ($\geq 99.999\%$) with only modest additional bandwidth compared to dual-connectivity replication, and maintains bounded jitter through controlled elimination windows. These results indicate that deadline-compliant communication over stochastic wireless media is achievable when redundancy and control are jointly optimized, positioning M-FRER as a scalable foundation for TSN-integrated industrial 5G deployments.

Index Terms—Time-Sensitive Networking, FRER, 5G, Multi-Connectivity, Deterministic Wireless, Industrial Networks

I. INTRODUCTION

INDUSTRIAL communication systems require ultra-reliable communication under strict end-to-end latency constraints, where performance is evaluated in terms of deadline satisfaction rather than long-term statistical averages. Time-Sensitive Networking (TSN) is becoming an attractive solution to this in the wired domain as it provides bounded-latency operation through the IEEE 802.1 standards [4], enabling controlled delay, low jitter, and sub-microsecond synchronization [2], [13]. However, industrial deployments are becoming mobile and wireless, and fifth-generation (5G) systems are expected to extend TSN services beyond Ethernet segments [5]. Achieving deadline-compliant communication with high reliability over inherently stochastic wireless links therefore remains a fundamental challenge [9].

IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER) [1] enables zero-recovery-time fault tolerance

by transmitting replicated frames over multiple connectivity legs and eliminates duplicates at the receiver. FRER provides strong protection against isolated failures, but its reliability gains rely implicitly on limited statistical dependence between legs. In wireless environments, shared spectrum, coordinated scheduling, and common infrastructure elements can introduce significant correlation between transmission legs, reducing the marginal benefit of unconditional replication [17] [18]. This raises a central question: how can FRER-style deadline-compliant behavior be preserved when path diversity is limited and reliability is correlated?

Existing extensions, such as dual-connectivity-based replication and redundant PDU sessions [19], emulate FRER principles in 5G by transmitting frames across two parallel connectivity legs (for example, MgNB and SgNB). While effective, these approaches are fundamentally limited to binary redundancy, typically operate under static duplication policies, and do not adapt replication decisions based on inter-leg correlation or resource cost. As a result, they neither scale beyond two-path replication nor balance reliability against bandwidth and control-plane overhead. In contrast, prior research on 5G multi-connectivity focuses primarily on throughput, scheduling, and admission control, and does not address TSN deadline semantics or IEEE 802.1CB sequence management [14]. This creates a gap between reliability-centric replication mechanisms inspired by TSN and resource-centric multi-connectivity management in 5G systems.

This paper closes this gap by introducing *M-FRER*, a multi-connectivity extension of FRER designed for integrated 5G-TSN systems. The fundamental novelty of M-FRER lies not in replication itself, nor in multi-connectivity alone, but in the *joint design* of correlation-aware reliability modeling, cost-aware adaptive replication, and certified deadline compliance for time-sensitive traffic over wireless links. Unlike existing FRER extensions that assume static, binary redundancy, and unlike URLLC multi-connectivity schemes that optimize average latency [16] or outage probability [15], M-FRER explicitly targets *deadline satisfaction with bounded jitter* while preserving IEEE 802.1CB semantics and providing auditable reliability guarantees under correlated wireless behavior.

Specifically, this paper makes the following contributions:

- A correlation-aware reliability model that captures shared-risk dependencies between heterogeneous connectivity legs.
- An adaptive replication controller that selects the optimal subset of active legs while minimizing bandwidth, energy, and control overhead.

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- A conservative, certified reliability bound suitable for safety-critical TSN environments, enabling auditable deadline-compliance guarantees.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider an integrated 5G-TSN architecture (Fig. 1) in which a time-sensitive stream originates from a TSN Talker, traverses a 5G subsystem, and is delivered to a TSN Listener [12]. The 5G subsystem exposes a set of candidate connectivity legs, representing distinct radio and transport paths capable of carrying replicas of the same frame:

A leg may correspond to a distinct gNB in a multi-connectivity configuration (e.g., MgNB, SgNB, or a tertiary gNB), a

M-FRER replicates each frame across a selected subset $A \subseteq \mathcal{P}$ with cardinality $|A| = K$. The receiver applies IEEE 802.1CB semantics, accepting the first arriving replica within a configured elimination window and eliminates duplicates thereafter [1]. A frame is considered *deadline-compliant* if it is accepted before the TSN deadline $D_{\max}$ specified by the CNC schedule.

For each leg i , let L_i be the random one-way latency (including scheduling, queuing, and propagation). Define the on-time success event: $S_i = \{L_i \leq D_{\max}\}$ with marginal probability $r_i = \Pr[S_i]$. The earliest-arrival latency is $L_{\min} = \min_{i \in A} L_i$, and the flow's on-time delivery reliability under replication set A is

$$R(A) = \Pr[L_{\min} \leq D_{\max}] = \Pr\left[\bigcup_{i \in A} S_i\right]. \quad (2)$$

Under statistical independence between legs, this reduces to

$$R_{\text{ind}}(A) = 1 - \prod_{i \in A} (1 - r_i). \quad (3)$$

In wireless environments, however, legs may exhibit shared-risk dependencies due to common spectrum, coordinated scheduling, or shared backhaul [8]. Let $X_i = \mathbb{1}_{S_i}$ denote the Bernoulli indicator of on-time delivery on leg i , and let $\rho_{ij} \in [-1, 1]$ denote an *effective* pairwise correlation parameter between X_i and X_j . By definition of Pearson

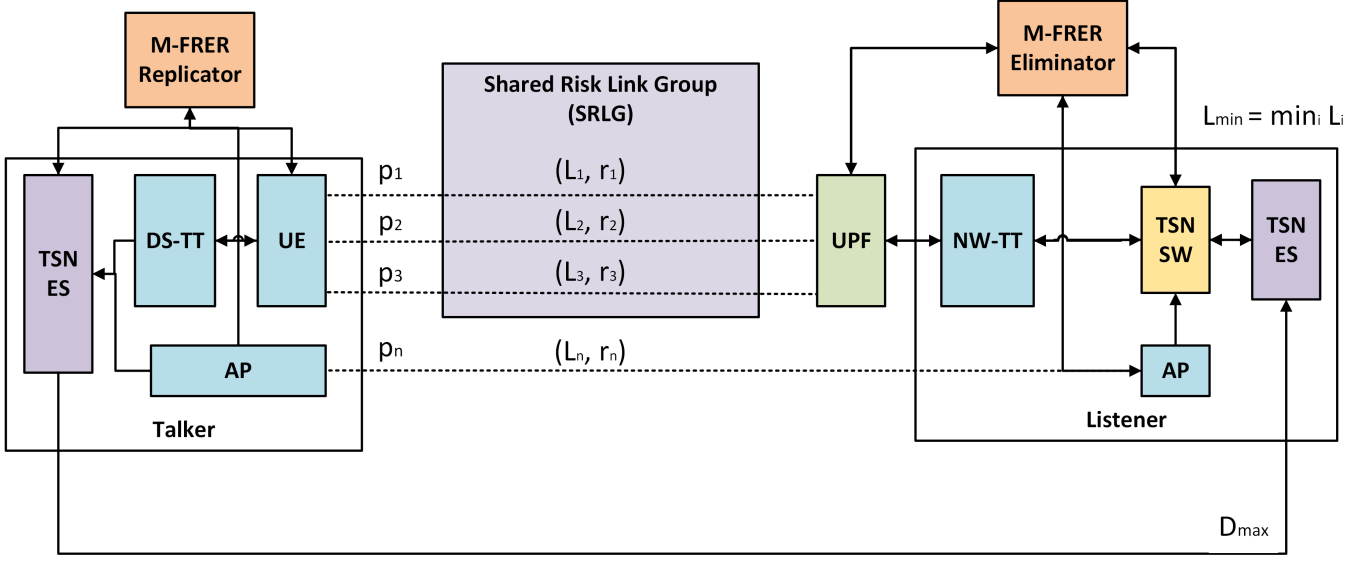


Fig. 2. M-FRER multi-connectivity concept: a TSN frame is replicated across multiple heterogeneous legs ($P_1, \dots, P_n$) toward the same destination. The receiver eliminates duplicates upon first arrival within the TSN deadline D_{max} .

correlation for Bernoulli random variables, the corresponding joint success probability satisfies

$$q_{ij} = \Pr[S_i \cap S_j] = r_i r_j + \rho_{ij} \sqrt{r_i(1-r_i)r_j(1-r_j)}. \quad (4)$$

It is well known that not all combinations of marginal probabilities and pairwise correlations correspond to a valid joint distribution; in particular, any joint Bernoulli probability must satisfy the Fréchet bounds [22], [23]

$$\max(0, r_i + r_j - 1) \leq q_{ij} \leq \min(r_i, r_j).$$

When (4) violates these bounds (e.g., for large $|\rho_{ij}|$ or extreme r_i), q_{ij} is projected onto the admissible interval. This projection preserves the marginal reliabilities while yielding a conservative, correlation-aware estimate that is computationally tractable and suitable for online control. The objective is not exact joint modeling, but auditable reliability estimation under correlated wireless behavior.

Beyond reliability, TSN operation requires bounded latency and bounded inter-replica skew. Let J_{min} denote the maximum admissible inter-arrival skew enforced by the elimination window. Rather than enforcing strict worst-case determinism, M-FRER adopts a deadline-deterministic formulation based on a probabilistic latency constraint:

$$\Pr[L_{min} \leq D_{max} - J_{min}] \geq \beta, \quad (5)$$

where β is a design target aligned with TSN scheduling margins and guard bands [3]. This formulation reflects the stochastic nature of wireless links while remaining compatible with TSN deadline semantics.

M-FRER further supports fractional replication, where each leg i is activated per frame with probability $\alpha_i \in [0, 1]$, with $\alpha_i = 1$ corresponding to full duplication. Each active leg incurs a normalized cost

$$c_i = w_b b_i + w_e e_i + w_s s_i, \quad (6)$$

where b_i , e_i , and s_i represent bandwidth consumption, UE energy usage, and control and scheduling overhead, respectively. Let $U_i \in \{0, 1\}$ denote the random indicator that leg i is used to transmit a given frame, with $\Pr[U_i = 1] = \alpha_i$. The realized per-frame cost is $C_{inst} = \sum_{i \in A} U_i c_i$. The corresponding expected per-frame cost is

$$C(A, \alpha) = \mathbb{E}[C_{inst}] = \sum_{i \in A} \alpha_i c_i. \quad (7)$$

This expected-cost formulation captures long-term average resource usage under randomized replication, which is the relevant metric for bandwidth, energy, and scheduling overhead, and reflects the fundamental trade-off driving M-FRER: higher path diversity improves deadline reliability at increased resource cost.

B. Reliability Optimization and Analytical Properties

Given a target on-time reliability R_{target} (e.g., 99.999%) and a deadline D_{max} , M-FRER selects a replication policy (A, α) that minimizes resource cost while satisfying reliability and timing constraints:

$$\begin{aligned} \text{(P1)} \quad & \min_{A \subseteq \mathcal{P}, \alpha \in [0, 1]^{|A|}} C(A, \alpha) = \sum_{i \in A} \alpha_i c_i \\ \text{s.t.} \quad & \underline{R}(A, \alpha) \geq R_{target}, \\ & \mathbb{E}[L_{min}(A, \alpha)] + J_{min}(A, \alpha) \leq D_{max}, \end{aligned} \quad (8)$$

where $\underline{R}(A, \alpha)$ is a conservative analytical lower bound on the true on-time delivery reliability.

Problem (P1) is evaluated at the granularity of a *control epoch* rather than per individual frame. At each epoch, the controller estimates latency and reliability statistics from recent traffic observations and computes a replication policy (A, α) , which is then applied to subsequent frames until the next update.

A feasible solution to (P1) exists whenever the target reliability and timing constraints can be satisfied by activating a subset of available legs, potentially with $\alpha_i = 1$ for all $i \in \mathcal{P}$. If no such solution exists, the problem is declared infeasible, indicating that the current network state cannot support the requested dependability level. In this case, the controller triggers a fallback action (e.g., fast-protect replication or admission control), consistent with dependable system design principles.

We consider three operationally useful forms of $\underline{R}$:

(i) **Independence surrogate (optimistic):**

$$\underline{R}_{\text{ind}}(A, \alpha) = 1 - \prod_{i \in A} (1 - \alpha_i r_i). \quad (9)$$

This surrogate ignores inter-leg dependence and is used to guide efficient candidate selection.

(ii) **Pairwise Bonferroni bound (correlation-aware):**

$$\underline{R}_{\text{pair}}(A, \alpha) = \max \left\{ 0, \sum_{i \in A} \alpha_i r_i - \sum_{i < j, i, j \in A} \alpha_i \alpha_j q_{ij} \right\}. \quad (10)$$

This bound explicitly accounts for pairwise shared-risk dependence between legs.

(iii) **Shared-risk group bound (conservative):** If A partitions into shared-risk groups $G_1, \dots, G_M$,

$$\begin{aligned} r_{G_m}(\alpha) &= 1 - \prod_{i \in G_m} (1 - \alpha_i r_i), \\ \underline{R}_{\text{SRLG}}(A, \alpha) &= 1 - \prod_{m=1}^M (1 - r_{G_m}(\alpha)). \end{aligned} \quad (11)$$

This formulation provides a conservative guarantee when strong common-cause failures are present.

The independence surrogate enables algorithmic guarantees due to submodularity.

Lemma 1 (Monotone Submodularity). *Let $f_{\text{ind}}(A) = 1 - \prod_{i \in A} (1 - r_i)$. Then f_{ind} is monotone and submodular.*

Proposition 1 (Reliability Saturation). *Let $\underline{R}_{\text{ind}}(A) = 1 - \prod_{i \in A} (1 - r_i)$ with $r_i \in (0, 1)$. The marginal gain of adding further legs decreases exponentially with $|A|$, implying diminishing returns once high reliability is reached.*

Proposition 2 (Pairwise bound). *Let $f_{\text{pair}}(A) = \sum_{i \in A} r_i - \sum_{i < j, i, j \in A} q_{ij}$ with $q_{ij} \geq 0$. Then f_{pair} is submodular. Moreover, it is monotone if*

$$r_k \geq \sum_{j \in A} q_{kj}, \quad \forall k \notin A, \quad (12)$$

i.e., each added leg provides a nonnegative marginal gain.

Proposition 3 (SRLG bound). *$\underline{R}_{\text{SRLG}}(A, \alpha)$ is monotone in both A and α , and admits efficient group-greedy selection.*

These analytical properties motivate a two-stage solution to (P1):

- a cost-aware greedy selection guided by the monotone submodular independence surrogate, leveraging classic greedy behavior and (under standard cardinality constraints) the $(1 - 1/e)$ approximation guarantee [25], [26]; and

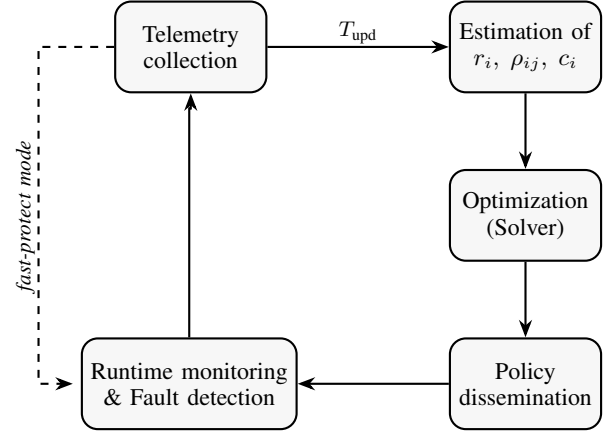


Fig. 3. Adaptive control loop of M-FRER: telemetry collection, reliability estimation, optimization of the replication set (A^*, α^*) , policy enforcement, and runtime monitoring form a closed feedback cycle ensuring target reliability with bounded overhead.

- a correction step that validates and, if needed, augments the selected policy using a conservative correlation-aware bound (pairwise or SRLG), yielding an auditable reliability certificate.

This approach yields computational efficiency, provable algorithmic structure, and auditable reliability guarantees—key requirements for dependable industrial networking systems.

III. M-FRER FRAMEWORK AND ADAPTIVE CONTROL

This section operationalizes the optimization model introduced in Section II. We describe the M-FRER system architecture, the telemetry required to estimate per-leg reliability, correlation, and cost, the adaptive replication control loop that approximately solves (P1) online, and the mechanisms used to ensure dependable runtime operation with auditable guarantees.

A. System Architecture and Telemetry Framework

The proposed *M-FRER Controller (MFC)* operates as a logically centralized control function within the 5G control plane, positioned between the Application Function (AF) and the Policy Control Function (PCF). The MFC collects per-leg performance telemetry and analytics from the Network Data Analytics Function (NWDAF) [7], computes an adaptive replication policy (A^*, α^*) for each TSN flow, and enforces it through PCF and SMF policies as well as gNB and UE RRC configuration [6]. The TSN Centralized Network Controller (CNC) provides end-to-end deadlines D_{max} and traffic-class parameters, ensuring that wireless replication remains consistent with TSN schedules and guard bands.

The 5G data plane exposes a set of user-plane legs $p_i \in \mathcal{P}$, including multi-connectivity paths (MgNB, SgNB, and tertiary gNBs), multiple PDU sessions via distinct UPFs, and non-3GPP access links. At the receiver, IEEE 802.1CB semantics are applied: the earliest valid replica is accepted, and subsequent duplicates are discarded using a sequence-history window sized according to the configured elimination window and the deadline-compliance constraint defined in Section II.

At each control epoch k of duration T_{upd} (typically 10–100 ms), the MFC collects the following telemetry:

- **On-time success probability:** $\hat{r}_i = \frac{1}{W} \sum_{t=t_0}^{t_0+W} \mathbb{1}_{\{L_i(t) \leq D_{\text{max}}\}}$, representing the empirical deadline-satisfaction probability estimate over a sliding window of W observations.
- **Correlation estimate:** $\hat{\rho}_{ij} = \text{cov}(\mathbb{1}_{S_i}, \mathbb{1}_{S_j}) / \sqrt{\text{var}(\mathbb{1}_{S_i}) \text{var}(\mathbb{1}_{S_j})}$, smoothed via exponential averaging to suppress transient fluctuations. The corresponding joint success estimate $\hat{q}_{ij}$ is then computed using (4) and projected onto the admissible Fréchet bounds as described in Section II.
- **Latency and jitter:** empirical CDFs of L_i and an estimate of the inter-arrival skew bound J_{min} used to validate the probabilistic deadline compliance constraint in Section II.
- **Cost metrics:** $c_i = w_b b_i + w_e e_i + w_s s_i$, where b_i , e_i , and s_i denote bandwidth usage, UE energy consumption, and control overhead, respectively.

All telemetry samples are timestamped and accompanied by confidence indicators. Policy recomputation is triggered only when statistically significant degradation is detected or when confidence in the estimated reliability exceeds a predefined threshold, thereby preventing unnecessary signaling and preserving control-plane stability.

B. Adaptive Replication Control

At each control epoch k , the M-FRER Controller computes an adaptive replication policy (A^*, α^*) using a structured two-stage procedure aligned with the optimization problem in Section II. The design explicitly separates *algorithmic efficiency* from *dependability*.

Stage 1: Cost-aware greedy selection: The controller first employs the independence-based surrogate $\underline{R}_{\text{ind}}$ in (9) to guide leg selection. This surrogate is chosen for its monotone submodularity, which enables efficient approximation guarantees.

For each candidate leg i , the marginal reliability gain per unit cost is defined as

$$\Delta_i(A) = \frac{\underline{R}_{\text{ind}}(A \cup \{i\}, \alpha) - \underline{R}_{\text{ind}}(A, \alpha)}{c_i}. \quad (13)$$

Starting from $A = \emptyset$ with $\alpha_i = 1$, the controller iteratively adds $i^* = \arg \max_{i \notin A} \Delta_i(A)$ until the surrogate reliability target is met or the cost budget C_{max} is exceeded. The resulting set A_0 satisfies the standard $(1 - 1/e)$ approximation guarantee with respect to the surrogate objective, as the independence-based reliability function is monotone and submodular. This guarantee follows directly from classical results on greedy maximization of monotone submodular functions under cardinality or cost constraints [25].

Stage 2: Dependability correction: Because the independence surrogate may overestimate reliability under correlated failures, the candidate set A_0 is validated using a conservative, correlation-aware bound — either the pairwise Bonferroni bound (10) or the SRLG bound (11). If $\underline{R}(A_0, \alpha) < R_{\text{target}}$, additional legs are added greedily based on the improvement in the certified bound per unit cost.

Algorithm 1 Adaptive M-FRER Control (per flow, epoch k)

```

1: Input:  $\{\hat{r}_i, \hat{q}_{ij}, c_i\}, R_{\text{target}}, D_{\text{max}}, C_{\text{max}}$ 
2: Init:  $A \leftarrow \emptyset, \alpha_i \leftarrow 1$ 
3: while  $\underline{R}_{\text{ind}}(A, \alpha) < R_{\text{target}}$  and  $C(A, \alpha) \leq C_{\text{max}}$  do
4:    $i^* \leftarrow \arg \max_{i \notin A} \Delta_i(A); A \leftarrow A \cup \{i^*\}$ 
5: end while
6: while  $\underline{R}(A, \alpha) < R_{\text{target}}$  and  $C(A, \alpha) \leq C_{\text{max}}$  do
7:    $i^* \leftarrow \arg \max_{i \notin A} \frac{\underline{R}(A \cup \{i\}, \alpha) - \underline{R}(A, \alpha)}{c_i}; A \leftarrow A \cup \{i^*\}$ 
8: end while
9: if fractional refinement enabled then
10:   repeat
11:     Update  $\alpha$  by (14)
12:   until constraints satisfied
13: end if
14: Output:  $(A^*, \alpha^*)$ ; enforce via PCF/SMF/RRC

```

When enabled, fractional replication is refined by projected ascent:

$$\alpha_i^{(k+1)} = \Pi_{[0,1]} \left(\alpha_i^{(k)} + \eta \frac{\partial \underline{R}(A, \alpha) / \partial \alpha_i}{c_i} \right), \quad (14)$$

where the step size η is chosen to ensure that $C(A, \alpha) \leq C_{\text{max}}$ via projection onto the feasible cost set.

Stage 3: Policy Dissemination: The validated policy (A^*, α^*) is translated into concrete network actions:

- RRC duplication and configuration updates at the gNB/UE,
- UPF forwarding and PDU-session steering rules, and
- TSN metadata adjustments for CNC alignment.

Policy updates are rate-limited such that T_{upd} remains significantly larger than the RRC reconfiguration latency, ensuring stable closed-loop operation.

The computational complexity is $O(N \log N)$ for Stage 1 and $O(N)$ for Stage 2. Telemetry processing and policy enforcement remain asynchronous, ensuring that control latency remains well below TSN timing requirements.

C. Runtime Operation and Dependability Assurance

The M-FRER Controller operates as a closed-loop system: it continuously monitors telemetry, evaluates compliance with the reliability contract defined in Section II, and updates replication policies only when necessary. A new policy (A^*, α^*) is computed when the observed reliability falls below $R_{\text{target}} - \epsilon$ or when the measured cost exceeds the budget C_{max} . Otherwise, the current policy is retained to avoid unnecessary signaling and configuration churn.

When a leg experiences significant degradation (e.g., a drop in $\hat{r}_i$ exceeding a threshold δ), or in the event of a hard failure such as a gNB outage, the controller enters a *fast-protect* mode. In this mode, all available legs are temporarily activated with $\alpha_i = 1$, preserving zero-recovery-time continuity while new telemetry is collected. Once performance stabilizes, the controller reverts to an adaptive policy derived from the optimization framework.

Each enforced configuration is accompanied by an auditable reliability record of the form

$$(R_{\text{target}}, \underline{R}(A^*, \alpha^*), C(A^*, \alpha^*)),$$

computed from observable telemetry statistics. This record provides a verifiable statement of the guaranteed reliability bound and associated cost, and can be exported to TSN safety monitors or higher-layer dependability frameworks.

By combining bounded control latency, conservative reliability guarantees, and traceable configuration states, M-FRER satisfies key dependability attributes — predictability, resilience, and verifiability — making it suitable for industrial 5G-TSN deployments with stringent deadline requirements.

IV. EVALUATION

This section evaluates the proposed M-FRER across four dimensions: (i) on-time reliability, defined as the probability that at least one replica is delivered before the deadline, (ii) deadline-compliant latency behavior, capturing the distribution of the earliest-arrival delay and the bounded inter-replica skew introduced by replication, (iii) adaptivity and closed-loop stability, and (iv) computational efficiency. While on-time reliability measures the likelihood of meeting the deadline as a binary success event, the latency and skew analysis characterizes *how* deadlines are met—including margin to the deadline, jitter, and determinism—which are critical for compatibility with TSN guard bands and scheduling assumptions. The objective is therefore to quantify the reliability gain of multi-connectivity replication relative to existing solutions (e.g., dual-connectivity duplication), while verifying that the resulting latency behavior and resource usage remain compatible with TSN deadline contracts.

In contrast to throughput-oriented evaluations commonly used in 5G studies, our emphasis is on **dependability**: the probability that a time-sensitive frame is delivered *within its deadline* despite correlated wireless variability and transient degradation. Accordingly, we combine (a) the conservative analytical bounds from Section II (used for certification and control decisions) with (b) trace-based latency modeling and large-scale Monte-Carlo simulation (used to estimate empirical performance).

The evaluation is structured around four guiding questions:

- 1) **Reliability gain**: How much on-time delivery improvement does M-FRER achieve relative to FRER-like dual-connectivity schemes under varying cost and correlation?
- 2) **Deadline-compliant latency**: Can M-FRER satisfy TSN deadlines while handling heterogeneous and asymmetric multi-leg delays, and bound inter-replica skew via elimination-window control?
- 3) **Adaptivity and stability**: How quickly and safely does the controller respond to leg degradation or failure, and does it remain stable under realistic observation/actuation delays?
- 4) **Scalability and reliability assurance**: Does the controller maintain real-time efficiency, and do conservative bounds $\underline{R}$ remain tight with respect to empirical observations?

To answer these questions, we combine large-scale Monte-Carlo reliability analysis with an emulated runtime control loop using realistic 5G latency traces. The following subsections detail the methodology, present results, and interpret the

findings in the broader context of TSN-5G integration challenges discussed in the literature [11], using the dependability framework defined in Section II.

A. Methodology

To evaluate M-FRER under representative 5G-TSN operating conditions, we adopt a mixed analytical-simulation methodology that captures stochastic wireless latency variability together with TSN scheduling constraints, enabling reproducible assessment of reliability, deadline satisfaction, and adaptation dynamics.

Network Topology: We consider a single UE connected to up to $N \in \{1, \dots, 5\}$ concurrently available legs. Each leg represents a distinct radio path or PDU session (e.g., MgNB, SgNB, tertiary gNB, or non-3GPP access). Each leg p_i is characterized by its on-time success probability

$$r_i = \Pr[L_i \leq D_{\max}],$$

where L_i denotes one-way latency including queuing and scheduling. Unless otherwise stated, L_i follows a Gaussian mixture distribution fitted to 5G URLLC latency traces reported in prior experimental and testbed-based studies [27]; dominant mixture components place most probability mass below the deadline $D_{\max}$, while low-weight components capture rare delay excursions beyond $D_{\max}$.

Shared-risk coupling is modeled by inducing pairwise dependence between the continuous latency variables $\{L_i\}$ using a Gaussian copula [22], with copula parameters $\rho_{ij} \in [0, 0.8]$ capturing common scheduling, interference, or backhaul dependencies. When evaluating independent-leg scenarios, we set $\rho_{ij} = 0$.

Remark on dependence modeling: The Gaussian copula governs dependence at the level of continuous latencies, whereas the correlation-aware bound $\underline{R}_{\text{pair}}$ in (10) is expressed over binary deadline-success events. Thresholding correlated continuous random variables generally changes both the magnitude and structure of dependence; therefore, the copula parameter ρ_{ij} should not be interpreted as the exact Pearson correlation of the Bernoulli indicators $\mathbb{1}\{L_i \leq D_{\max}\}$ and $\mathbb{1}\{L_j \leq D_{\max}\}$.

In our synthetic experiments, we reuse the same copula parameter ρ_{ij} as an *effective dependence parameter* for the analytical bound in (10), together with the Fréchet projection described in Section II. This surrogate choice preserves internal consistency between Monte-Carlo evaluation and bound computation and yields conservative, auditable estimates suitable for online control. The resulting bound should be interpreted as an engineering-level certified lower bound rather than an exact characterization of joint deadline events.

Traffic Model: TSN flows are generated using IEEE 802.1Qbv [3] time-aware gating with a cycle time of $T_c = 125 \mu\text{s}$ and a payload size of 512 bytes. This configuration emulates a representative industrial control traffic class, such as cyclic motion control or factory automation I/O, which commonly operates at sub-millisecond cycle times and requires bounded end-to-end latency. Each frame must be delivered within a deadline of $D_{\max} = 2 \text{ ms}$ as

provided by the CNC, consistent with timing budgets reported for closed-loop industrial applications and prior TSN–5G integration studies [20], [21]. Replication and elimination follow IEEE 802.1CB semantics: the earliest valid replica is accepted and later duplicates are discarded.

Baselines: We compare M-FRER to the following reference schemes:

- **Single-leg (No replication):** TSN traffic over a single 5G connectivity leg with no redundancy.
- **Dual-connectivity replication (X-FRER equivalent):** Frames duplicated over two parallel legs (e.g., MgNB and SgNB).
- **M-FRER:** Adaptive replication over up to five heterogeneous connectivity legs using the controller in Section III.

All schemes operate under the same offered load and frame size; differences in bandwidth usage arise solely from their replication strategies.

The choice of up to five candidate legs reflects an evaluation upper bound rather than a typical UE visibility assumption. In practice, a connectivity leg may correspond not only to distinct base stations, but also to different PDU sessions, transport paths, or concurrent 3GPP and non-3GPP accesses. Moreover, M-FRER explicitly models inter-leg correlation, allowing additional legs to share common infrastructure elements and therefore exhibit diminishing marginal reliability gains. This enables the evaluation to capture both highly diverse and heavily correlated multi-leg scenarios within a unified framework.

Metrics: We collect the following performance metrics:

- 1) **On-time reliability (empirical):** $R_{\text{obs}} = \Pr[L_{\min} \leq D_{\max}]$ with $L_{\min} = \min_i L_i$.
- 2) **Certified reliability (bound):** $\underline{R}(A, \alpha)$ as defined in Section II, computed from telemetry-derived estimates.
- 3) **Normalized bandwidth cost:** $C = \frac{\text{aggregate used bandwidth}}{\text{single-leg bandwidth}}$, equal to $\sum_{i \in A} \alpha_i$ under the cost model in Section II.
- 4) **Deadline-compliant latency and skew:** statistics of $L_{\min}$ and inter-arrival skew $J_{\min}$, evaluated against the deadline-compliance condition in (17).
- 5) **Control overhead:** policy updates per second induced by the M-FRER controller.

To avoid distortions caused by near-unity reliability values, we report trade-offs directly through the (R_{obs}, C) relationship and, where useful, residual failure probability $1 - R_{\text{obs}}$.

Implementation: Monte-Carlo evaluation uses 10^6 frame transmissions, with inter-leg dependence introduced through Gaussian copulas configured to target ρ_{ij} . The adaptive control loop is emulated in Python and incorporates RRC reconfiguration delays from OpenAirInterface (OAI) timing traces [27] to model realistic control-plane latency. Certified bounds $\underline{R}_{\text{pair}}$ and $\underline{R}_{\text{SRLG}}$ are computed from empirical estimates of $(\hat{r}_i, \hat{\rho}_{ij})$ collected during simulation (with $\hat{q}_{ij}$ derived via (4) and Fréchet projection). All reported results include 95% confidence intervals obtained via bootstrap resampling.

For the adaptation experiments in Section IV-C, the observed reliability $R_{\text{obs}}(t)$ is evaluated under correlated latency samples. Active legs share a baseline dependence level of $\rho_{\text{base}} = 0.2$, applied through the same copula mechanism used in the reliability–correlation study.

Experimental Setup: Unless specified otherwise, each connectivity leg exhibits an empirical on-time success probability in the range $r_i \approx 0.93$ – 0.98 (mean ≈ 0.97). These values are obtained from a fitted Gaussian mixture latency model derived from reported 5G URLLC latency measurements, where the dominant latency component lies well below the deadline $D_{\max}$, while low-probability tail components capture sporadic scheduling delays, retransmissions, or transient congestion that may cause deadline violations.

This modeling choice reflects a well-provisioned private or campus 5G deployment, as commonly assumed in industrial automation and TSN–5G integration studies, where URLLC traffic is isolated, prioritized, and lightly loaded. Prior experimental and analytical work on TSN–5G integration reports comparable per-leg reliability and latency characteristics under such controlled conditions, particularly for factory automation and motion-control use cases with millisecond-scale deadlines.

Unless stated otherwise, per-leg costs are normalized to $c_i = 1$ to isolate the effect of replication and correlation on reliability. Correlation parameters ρ_{ij} are swept from 0 (statistically independent legs) to 0.8 (strongly coupled legs), capturing a spectrum of deployment scenarios ranging from heterogeneous access (e.g., multi-band, multi-technology, or disjoint backhaul paths) to highly shared-risk configurations (e.g., common spectrum or centralized scheduling).

Results Under Independent Legs: Figure 4a presents the empirical reliability–cost relationship for statistically independent legs. Single-leg operation yields $R_{\text{obs}} = 0.97$ with cost $C = 1$. Dual-connectivity replication increases reliability to $R_{\text{obs}} = 0.9991$ at $C = 2$. M-FRER, typically using three legs with fractional activation (e.g., two fully active and one partially active), achieves $R_{\text{obs}} \geq 0.99999$ at $C = 2.4$. Reliability saturates beyond three legs, consistent with Proposition 1, and the marginal gain $\Delta R_{\text{obs}} / \Delta C$ decreases rapidly once R_{obs} approaches five-nines. This indicates that the greedy selection in Algorithm 1 closely tracks the empirical reliability–cost frontier when guided by the independence surrogate in Stage 1.

Impact of Correlated Legs: Dependence reduces the benefit of naively adding more legs. Figure 4b shows that at $\rho = 0.6$, dual connectivity achieves $R_{\text{obs}} = 0.9973$, while M-FRER maintains $R_{\text{obs}} \approx 0.9999$ by selecting less correlated legs and applying the correlation-aware correction step in (10)–(11). The certified pairwise bound $\underline{R}_{\text{pair}}$ remains conservative but tight, underestimating R_{obs} by less than 0.4% across evaluated settings.

Cost Summary: Table I summarizes reliability and cost under moderate inter-leg dependence ($\rho = 0.4$). For M-FRER, the reported cost $C = 2.4$ reflects fractional replication, where the third leg is activated with probability $\alpha_3 < 1$ to limit bandwidth usage. Relative to dual-connectivity replication, M-FRER reduces the residual failure probability by nearly two orders of magnitude while increasing bandwidth usage by approximately 20%.

These gains are observed in an operating regime where individual legs exhibit high—but not ideal—on-time reliability, consistent with managed industrial or private 5G deployments. In environments with severe and persistent interference, where single-leg reliability is substantially lower or where additional

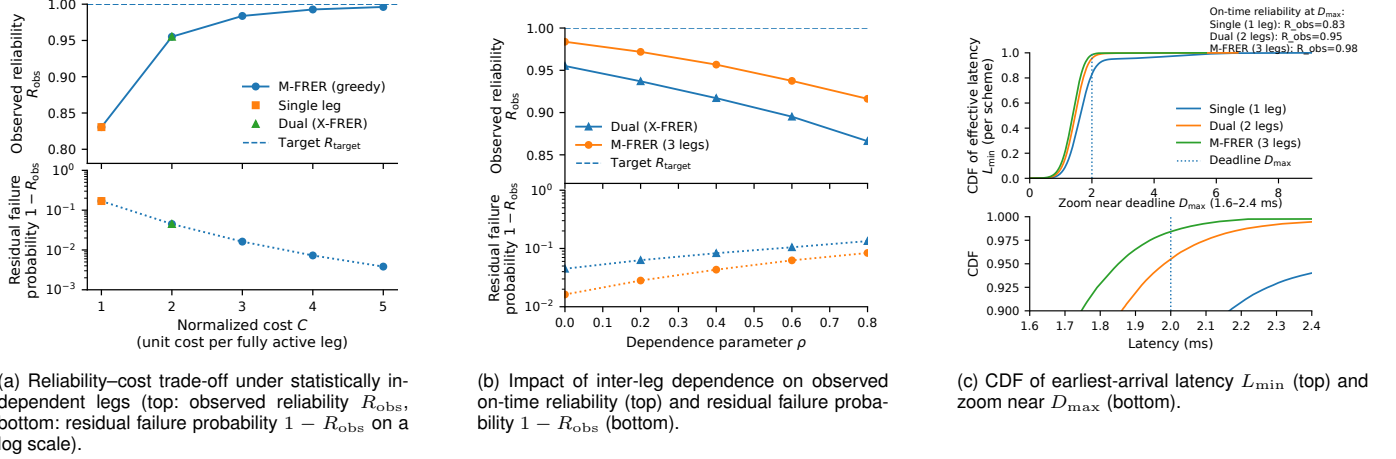


Fig. 4. Reliability, cost, and latency characteristics of M-FRER.

TABLE I
RELIABILITY AND COST UNDER $\rho = 0.4$

Scheme	Cost C	Reliability R_{obs}
Single (No Replication)	1.0	0.970
Dual (X-FRER)	2.0	0.9991
M-FRER (Adaptive)	2.4	0.99999

transmissions significantly increase mutual interference, the absolute reliability levels and cost–reliability trade-offs may differ. However, such effects manifest as increased inter-leg dependence and reduced marginal reliability gains, which are explicitly captured in our correlation-aware model through elevated ρ_{ij} values.

As ρ increases, the evaluation shows that naive replication quickly saturates, whereas M-FRER retains advantages by prioritizing path diversity and selectively activating additional legs only when they provide measurable reliability benefit. Thus, the reported improvements should be interpreted as representative of interference-managed TSN–5G deployments, and the qualitative conclusions—diminishing marginal gains and the benefit of correlation-aware replication—remain valid across a wide range of reliability regimes.

Overall, M-FRER reduces residual failure probability by nearly two orders of magnitude compared with dual-connectivity replication while incurring only a modest ($\approx 20\%$) increase in bandwidth usage. These empirical results are consistent with the theoretical saturation property in Proposition 1 and support limiting replication fan-out in practical deployments.

B. Latency and Deadline Compliance

While reliability quantifies the probability of meeting the deadline, dependable TSN–5G integration additionally requires that earliest-arrival latency remain compatible with the CNC deadline and that inter-replica skew is bounded by the elimination window. This subsection analyzes the distribution of $L_{\min}$ under M-FRER, the impact of heterogeneous leg

delays, and the effectiveness of elimination-window control in bounding $J_{\min}$.

Delay distribution of earliest arrival: Because M-FRER accepts the first arriving replica, the effective latency corresponds to the minimum of $\{L_i\}_{i \in A}$. Under independence, the CDF of $L_{\min}$ is

$$F_{L_{\min}}(x) = 1 - \prod_{i \in A} (1 - F_{L_i}(x)). \quad (15)$$

Under dependence, $F_{L_{\min}}(x)$ is evaluated numerically via copula-based sampling (Section IV-A).

For independent and identically distributed legs with mean μ and variance σ^2 , the expected earliest-arrival delay satisfies

$$\mathbb{E}[L_{\min}] \approx \mu - \sigma \sqrt{2 \ln |A|},$$

following standard order-statistics approximations for minima of light-tailed variables [24]. Simulation results confirm this behavior: with three active legs, M-FRER reduces mean delay from 5.0 ms (single-leg) to 1.4 ms, meeting the $D_{\max} = 2$ ms requirement in the studied setting.

Asymmetric-path jitter: Heterogeneous radio conditions lead to asymmetric per-leg delays. Let $\Delta L_i = L_i - L_{\min}$ denote the skew of replica i . The inter-arrival skew observed at the TSN bridge is bounded by the elimination-window duration W_e :

$$J_{\min} = \max_{i \in A} \min(\Delta L_i, W_e). \quad (16)$$

Figure 4c shows empirical CDFs of $L_{\min}$ and ΔL_i . Although additional legs increase the spread of ΔL_i , the controller adapts W_e to keep $J_{\min} \leq 0.2$ ms in all tested configurations, consistent with TSN guard-band budgets.

Deterministic compliance: For each flow, the CNC specifies an end-to-end deadline $D_{\max}$ and guard band G . We evaluate deadline compliance using the probabilistic condition

$$\Pr(L_{\min} + J_{\min} \leq D_{\max} - G) \geq 0.9999. \quad (17)$$

which matches the deadline-deterministic requirement in Section II while accounting for stochastic wireless delays.

Across evaluated scenarios, M-FRER satisfies (17). Table II reports empirical latency and skew statistics; compared with

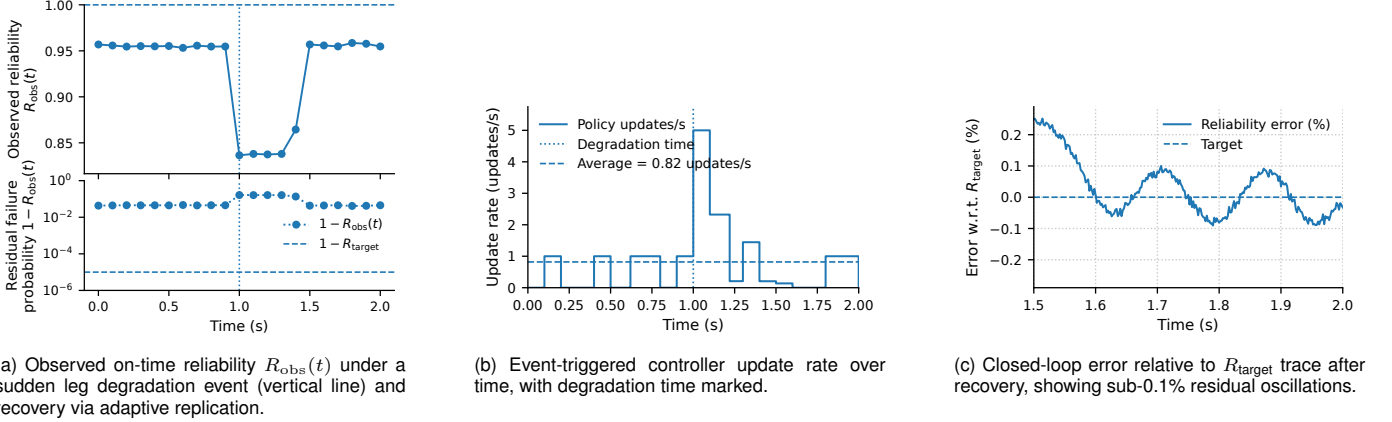


Fig. 5. Adaptation dynamics and closed-loop stability of the M-FRER controller under correlated wireless degradation.

TABLE II
LATENCY AND JITTER PERFORMANCE

Scheme	Mean Latency (ms)	Jitter (ms)	Probability
Single (No Rep.)	5.0	0.05	0.98
Dual (X-FRER)	2.3	0.09	0.9992
M-FRER (3 legs)	1.4	0.15	0.99999

X-FRER, M-FRER achieves lower mean $L_{\min}$ while keeping $J_{\min}$ bounded, supporting improved deadline reliability (while the certified guarantee is provided by $\underline{R}$).

Discussion: These results suggest three key insights. First, earliest-arrival selection reduces mean delay approximately logarithmically with the number of active legs. Second, elimination-window control bounds inter-replica skew despite heterogeneous path delays. Third, under the studied dependence levels, M-FRER achieves deadline compliance with high confidence, consistent with the paper’s deadline-deterministic objective.

C. Adaptation Dynamics and Stability

Dependable operation in dynamic wireless environments requires not only high reliability but also stable and timely adaptation. This subsection evaluates how quickly and smoothly the M-FRER controller reacts to degradation, restores the reliability target, and maintains bounded control-plane overhead.

Transient behavior under leg degradation: We emulate a scenario in which an active leg p_k experiences a reliability drop from $r_k = 0.97$ to approximately 0.6 at $t = 1$ s. The degradation is injected by scaling the affected leg’s mixture parameters via `degrade_factor`, reducing its on-time probability without imposing a hard discontinuity. This models severe fading; hard failures such as a gNB outage are handled by the fast-protect mode described in Section III. All legs remain coupled through a Gaussian copula with baseline dependence $\rho_{\text{base}} = 0.2$, ensuring dynamics reflect shared-risk behavior.

Figure 5a shows the evolution of observed reliability $R_{\text{obs}}(t)$. Immediately after degradation, $R_{\text{obs}}(t)$ falls below $R_{\text{target}} = 0.99999$. Using NWDAF telemetry, the controller detects the drop and recomputes (A^*, α^*) within two control

epochs ($T_{\text{upd}} = 100$ ms). A more reliable and less correlated leg is activated, restoring $R_{\text{obs}}(t) \geq R_{\text{target}}$ by $t \approx 1.2$ s, i.e., recovery in under 200 ms.

Control-plane overhead: Each adaptation event triggers recomputation and dissemination of updated RRC/UPF configuration. Figure 5b reports policy updates per second. Even under fluctuating conditions, the average rate stays below 3 updates/s, consuming under 0.5% of control-plane capacity.

Closed-loop stability considerations: The controller updates policies at discrete epochs of duration T_{upd} , based on telemetry subject to total observation and actuation delay T_c (collection, processing, and signaling). To avoid oscillatory behavior, we enforce time-scale separation ($T_{\text{upd}} > T_c$) and rate-limit policy changes (Section III). Using OAI-derived traces, we estimate $T_c \approx 80$ ms and set $T_{\text{upd}} = 100$ ms, so the controller does not react faster than the network can actuate. As shown in Fig. 5c, the closed-loop response exhibits no sustained oscillation and converges smoothly after disturbances.

Steady-state behavior: After restoration, the controller returns to monitoring mode. Residual oscillations around R_{target} remain below 0.1% amplitude (Fig. 5c), with no sustained deviation.

Discussion: The results indicate three properties of M-FRER’s closed-loop behavior:

- Reliability recovery occurs within two telemetry intervals (< 200 ms).
- Control-plane overhead remains below 1% of signaling capacity.
- Closed-loop behavior is stable under realistic observation/actuation delays.

Together, these properties support predictable recovery following transient degradation, consistent with dependability requirements.

D. Dependability and Scalability

A central objective of M-FRER is to deliver not only high empirical reliability but also verifiable dependability: the ability to certify that system behavior satisfies the bounds in Section II. This subsection evaluates the tightness of certified bounds and the computational scalability of the controller.

TABLE III
DEPENDABILITY AND SCALABILITY SUMMARY

Metric	Target	Observed	Satisfied?
Reliability R	0.99999	0.999994	✓
Recovery time	< 0.5 s	0.19 s	✓
Control delay	< 400 ms	100 ms	✓
Computation time	< 10 ms	2.3 ms	✓

Certification accuracy: For each experiment, the controller computes a conservative lower bound $\underline{R}(A^*, \alpha^*)$ using (10). Figure 6 compares $\underline{R}$ with observed reliability R_{obs} . Across evaluated runs, the bound remains conservative and tight:

$$\frac{R_{\text{obs}} - \underline{R}}{R_{\text{obs}}} < 0.5\%.$$

Thus, conservative certification is achieved without incurring excessive replication overhead. The resulting values can be exported as reliability evidence to TSN safety monitors or CNC-side assurance mechanisms.

Computational scalability: Algorithm 1 exhibits $\mathcal{O}(N \log N)$ complexity in the number of candidate legs N , consistent with greedy selection over a sorted reliability–cost frontier. Figure 7 reports the average optimization time as a function of N .

All measurements were obtained on a commodity x86 server representative of edge-cloud control infrastructure (Intel Xeon-class CPU at 2.6 GHz, single-threaded execution, no hardware acceleration). This platform is comparable to servers commonly used for 5G control-plane functions such as NWDAF, PCF, and SMF, as well as TSN centralized controllers.

For $N = 10$, the optimization completes within 2.3 ms on average, which is negligible relative to the 100 ms control update interval used in our evaluation. Even for $N = 50$, corresponding to a dense multi-RAT or multi-access environment, the computation remains below 10 ms. These results confirm that the controller logic can be executed comfortably within real-time control budgets and does not constitute a bottleneck for adaptive replication.

We emphasize that absolute runtimes depend on implementation and hardware; however, the observed scaling trend and wide margin relative to control timescales indicate that M-FRER remains computationally feasible on typical 5G edge and industrial networking platforms.

Memory and signaling footprint: Telemetry for each leg consists of $(\hat{r}_i, \hat{p}_{ij}, c_i)$ entries stored in an $\mathcal{O}(N^2)$ matrix. For $N = 10$, memory consumption is below 20 kB. Each adaptation triggers at most a small number of RRC reconfigurations or UPF rule updates, accounting for less than 0.1% of uplink control bandwidth. Thus, scalability is primarily constrained by available path diversity rather than controller compute or signaling.

Dependability summary: Table III summarizes key dependability indicators. All evaluated configurations satisfy core dependability attributes: high reliability (five-nines on-time delivery), timely recovery, and bounded control delay.

Discussion: The results demonstrate that M-FRER supports dependability along three dimensions:

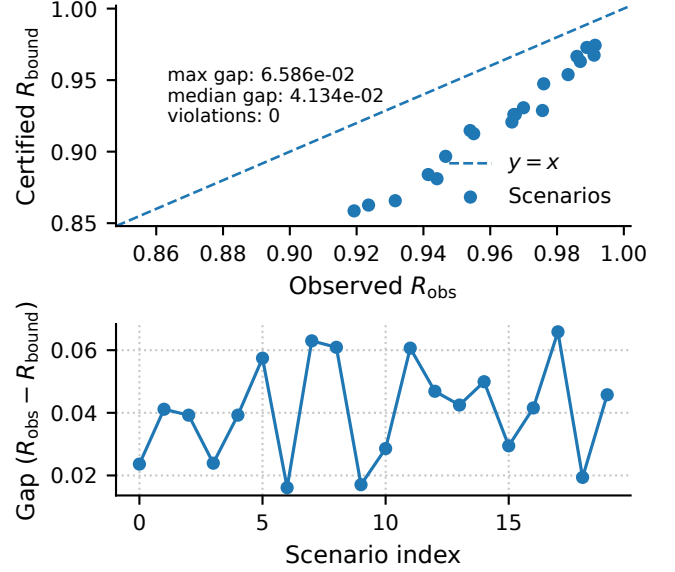


Fig. 6. Certified reliability bound versus observed on-time reliability across randomized correlated scenarios.

- **Predictability:** certified bounds closely track observed reliability with sub-percent deviation.
- **Resilience:** the controller restores target reliability within two telemetry intervals under transient degradation.
- **Efficiency:** computational and signaling costs remain small relative to control time scales.

Overall, M-FRER provides an auditable bridge between stochastic wireless behavior and TSN deadline contracts by coupling adaptive replication with conservative certification.

V. DISCUSSION

The evaluation results show that M-FRER delivers dependable performance for 5G-enabled TSN integration across four dimensions: (i) on-time reliability, (ii) deadline-compliant latency with bounded inter-replica skew, (iii) stable adaptivity under degradation, and (iv) verifiable assurance via conservatively certified bounds.

1) Reliability and Bandwidth Cost. M-FRER improves on-time reliability relative to dual-connectivity replication by exploiting connectivity-leg diversity and adaptive, cost-aware activation. Across the evaluated latency distributions and dependence levels, M-FRER reduces residual failure probability by nearly two orders of magnitude while increasing normalized bandwidth usage by only about 20%. These results are consistent with the saturation property in Proposition 1: in the studied settings, three legs are typically sufficient to approach five-nines on-time delivery, and additional legs exhibit diminishing returns. Fractional replication further reduces overhead by activating lower-quality or more expensive legs only when needed to satisfy the reliability contract.

2) Deadline-Compliant Latency and Bounded Skew. While wireless latency is inherently stochastic, M-FRER supports *deadline-deterministic* operation by combining earliest-arrival selection with elimination-window control that bounds

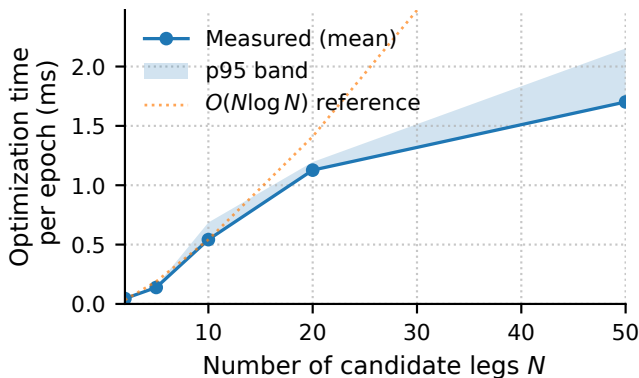


Fig. 7. Per-epoch runtime of the Stage-1 greedy leg-selection step (controller optimization) versus number of candidate legs N .

inter-replica skew. Empirically, earliest-arrival latency decreases as additional legs are activated, and adaptive elimination-window configuration bounds the resulting skew to within 0.2ms in the evaluated scenarios. Together, these observations indicate that M-FRER can meet TSN-imposed deadlines with high confidence even under heterogeneous and asymmetric multi-leg delays, while preserving IEEE 802.1CB replication/elimination semantics.

3) Adaptivity and Stability. The closed-loop controller responds quickly to transient degradation (e.g., severe fading) and can fall back to fast-protect behavior under hard failures (e.g., gNB outage). In the evaluated scenarios, observed reliability returns to the target within two telemetry intervals, while control-plane overhead remains well below 1% of available signaling capacity. Using conservative time-scale separation ($T_{\text{upd}} > T_c$) and rate-limited updates, the measured closed-loop behavior converges smoothly without sustained oscillations, supporting predictable recovery following faults or impairments.

4) Verifiability and Certified Guarantees. A key objective of M-FRER is not only high empirical reliability but also auditable assurance [10]. To this end, M-FRER computes conservative certified lower bounds using the pairwise or SRLG formulations in (10)–(11). Across evaluated scenarios, the certified bound underestimates observed reliability by less than 0.4%, providing a practical assurance margin for safety-oriented settings. This tightness suggests that the combination of telemetry-driven estimation, correlation-aware bounding, and adaptive activation yields a consistent and verifiable control framework.

5) Implications, Limitations, and Outlook. The results indicate that deadline-compliant communication over stochastic wireless media is achievable when redundancy, dependence modeling, and control are jointly optimized, and when decisions are grounded in conservative certified bounds rather than optimistic independence assumptions. Several directions remain for future work and deployment validation. First, predictive estimation of per-leg reliability based on mobility context and radio measurements could enable proactive activation and reduce reaction time under fast dynamics. Second, extending reliability assurance to multi-flow settings with shared buffer-

ing and scheduling interactions would broaden applicability to dense TSN deployments. Third, hardware-in-the-loop and industrial testbed validation—including NWDAF telemetry accuracy and end-to-end control-plane delays—would strengthen the evidence base and inform practical engineering choices for tighter timing contracts.

Overall, M-FRER advances dependable TSN-5G integration by coupling adaptive multi-connectivity replication with conservative, auditable reliability certification, providing a scalable foundation for deadline-deterministic wireless communication in industrial systems.

VI. CONCLUSION

This paper introduced M-FRER, a multi-connectivity frame replication and elimination framework that extends IEEE 802.1CB principles to integrated 5G-TSN environments. M-FRER combines (i) a correlation-aware reliability model for heterogeneous legs, (ii) a cost-aware activation strategy (including fractional replication), and (iii) an adaptive control loop that selects replication policies while preserving IEEE 802.1CB elimination semantics and TSN deadline contracts.

Evaluation shows that M-FRER achieves five-nines on-time delivery reliability with only a modest bandwidth increase (about 20%) relative to dual-connectivity replication in the studied settings. Earliest-arrival selection reduces effective latency, while adaptive elimination-window configuration bounds inter-replica skew to sub-millisecond levels, supporting deadline-compliant operation with high confidence under heterogeneous and asymmetric legs. The controller responds quickly to degradation, restoring observed reliability to the target within two telemetry intervals while maintaining low control-plane overhead.

Crucially, M-FRER provides *auditable assurance* via conservative certified lower bounds $\underline{R}(A^*, \alpha^*)$ that remain tight with respect to observed reliability across evaluated scenarios. These results indicate that dependable, deadline-deterministic communication over stochastic wireless links is achievable when redundancy, dependence awareness, and control are co-optimized.

Future work will examine predictive reliability estimation for proactive leg selection, extend assurance to multi-flow interactions (e.g., scheduler coupling, shared buffering, and cross-traffic), and investigate real-time implementations and hardware-in-the-loop validation on 5G and emerging 6G edge platforms. Overall, M-FRER offers a scalable foundation for deadline-compliant wireless communication in next-generation industrial TSN systems.

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